

The fate of rainfall in a mallee and heath vegetated catchment in southern Western Australia

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Abstract

Mallee, a shrub-eucalypt association, once covered large areas of the cereal growing land in Western Australia. The hydrologic consequences of land development have been more than a doubling of water yield and increased deep drainage beyond the plant-root zone. The latter has led to large areas of soil salinization.

The uncleared catchment studied was covered with mallee vegetation (65%) and a heath association (35%). Over 12 years of measurement, runoff from the catchment has averaged 0.025 mm per annum with a mean annual rainfall of 376 mm. Half of the runoff resulted from two major events totalling 0.15 mm. Since there was no evidence of groundwater accession in the catchment, the rainfall was balanced by evapotranspiration. Despite virtually no runoff from the whole catchment, considerable internal redistribution of surface water occurred, with localized overland flow in some areas as high as 7.7 mm from 30.9 mm of rainfall in one day.

Rainfall penetrated rapidly under the mallee vegetation and the rate of penetration observed could not be achieved solely through the soil even if there was saturated flow.

A typical stand of native mallee trees (Eucalyptus pileata and E. eremophila) in the catchment effectively redistributed 8% of the annual rain falling on the stand with 3% lost as interception and 5% going to stemflow. However, on an individual tree basis some 15% of the

rain falling on the canopy was lost as intercepted water and 25% ran down the stem. The stemflow caused saturated conditions around the bole of the mallee and dye tracing showed that the water penetrated the soil via the annular pathways of the soil-root interface.

Roots of mallee trees were found at 28 m depth and it is postulated that the mallees are adapted to the semi-arid environment by virtue of their ability to store water deep in the soil profile for use during the dry summer months.

Introduction

Mallee, a shrub-eucalypt association, once covered large areas of the now developed agricultural land in southern Western Australia. Replacement of the mallee with annual crops and pastures has resulted in significant salinization of soil in Western Australia (Smith 1962; Peck 1978). In 1979 it was estimated that some 1.75% (264 000 ha) of once productive agricultural land in the State was affected by secondary salinity (Henschke 1981).

The increase in deep drainage and subsequent accessions to the groundwater under the agricultural system is relatively small (Peck & Hurlé 1973). In a semi-arid region in South Australia, Allison and Hughes (1983) calculated that deep drainage increased by about 3 mm per year after *Eucalyptus* scrub was developed for cereal cropping. It is not only the deep drainage component of the hydrological cycle that is affected by clearing. Bligh (1981) showed that water yields at least doubled after clearing the native vegetation in southern Western Australia.

It is well documented that the plant canopy significantly redistributes rainfall before it reaches the soil surface. Specht (1957) reported that a *Banksia ornata*-*Xanthorrhoea australis* community intercepts and redistributes up to 70% of the incident rainfall. Mulga (*Acacia aneura*) has a canopy structure that causes 40%

of the rain falling on the canopy to flow down the stems (Slatyer 1965) and it has been suggested that this high stemflow is essential for the survival of mulga in arid environments (Pressland 1976). The high stemflow in this arid environment species contrasts with low stemflow of 0.6% for silverleaf ironbark (*E. melanophloia*) in a humid environment (Prebble & Stirk 1980).

The redistribution of rainfall by vegetation continues below ground level. Pressland (1976) found that stemflow from mulga was instrumental in storing water at depth, particularly from rainfall events around 75 mm. Larger events (> 160 mm) tended to mask the effect. Under the *Eucalyptus* scrub formation known as mallee, Allison and Hughes (1983) found recent water (post-1960) had penetrated to at least 12 m, whereas recent water was not found below 2.5 m under an adjacent cleared area.

Vegetation morphology also affects water distribution below the soil surface as shown by Specht (1957) who found that taller, and presumably deeper rooted species channelled water to greater depth than shorter, shallow rooted plants.

Often in analyses of hydrologic changes following clearing, it is tacitly assumed that the flow paths before and after clearing are geometrically similar and that only quantitative changes occur.

Allison and Hughes (1983) speculated that under mallee vegetation water was transported to depth via channels occupied by living roots. Removal of the mallee may make these channels inoperative, thus changing not only the quantity of water entering the soil, but also the geometry of its flow path.

The aim of this study was to describe the hydrology of a naturally vegetated catchment.

Site studied

Location

The study site (Fig. 1) was located at Wheelock's Catchment, 20 km ENE of Newdegate, Western Australia (33°06'S, 118°59'E) in gently undulating country with long, low angle slopes. Wheelock's Catchment is a sub-catchment draining north and eventually into Lake Stubbs, a playa lake immediately north of Newdegate township.

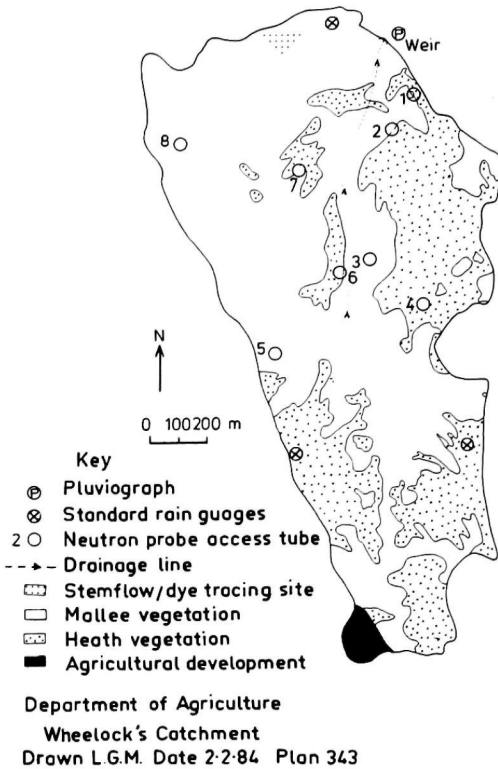


FIG. 1. Map of Wheelock's Catchment showing vegetation communities, neutron moisture meter access sites and dye tracing/stemflow measurement site.

The bed of Lake Stubbs is about 300 m above sea level.

Wheelock's Catchment, which is yet to be developed for agriculture, has an area of 265 ha, is covered in native vegetation (98.8%) and surrounded by land that has been developed for wheat and sheep farming for at least 20 years. The drainage line in the Catchment has a uniform gradient of 0.017.

Climate

The climate is Mediterranean, with an average rainfall (1972–82) of 376 mm, of which 60% falls during the 6 months from May to October. The highest daily rainfall recorded since 1972 was 52.5 mm on 27 February 1976. Annual class A pan evaporation for the area is 2200 mm.

Temperatures were not recorded on site but at the nearest station measuring temperatures. Lake Grace, some 80 km west, the January mean maximum and minimum temperatures

are 32°C and 14°C, respectively. For July, the respective temperatures are 15°C and 5°C.

Geology and soils

The geology of the Catchment is typical of the weathering profile of the granitic Archaean Shield. Superficial sandy soils are underlain by a silicified hardpan of cemented silty sandy clay from 1 to 4.5 m. Below the hardpan is an acidic (pH 5.5) pallid zone of kaolinitic clays with subangular quartz up to 5 mm diameter. The clays are predominantly micaceous below 17 m with hard, quartz-rich bands. Intergranular voids and solution channels were observed to 33 m in the one cored borehole south of Site 3 (Fig. 1). The bedrock is gently undulating about a depth of 50 m and there is no apparent relationship between the fresh bedrock configuration and present day topography. Seismic velocities of the bedrock ranged from 4.76 to 5.41 km s⁻¹ indicating little likelihood of jointing (Rowston 1982).

The laterite varies from pisolitic to massive and there is a small outcrop of massive laterite on the eastern boundary of the Catchment.

The soils of the area are hard alkaline yellow mottled soils (Dy 3.43 and Dy 3.42; Northcote 1974). Soils of the Catchment consist predominantly of either brown loamy sand or sandy gravel over pale brown to light yellowish brown mottled sandy clay. The A horizon varies from 0.05 to 0.30 m deep and has a neutral or slightly acid reaction. The mottled zone extends to the hardpan at 3 to 5 m and is alkaline with pH (1:5 soil:water) of 8.4–9.3. Large (5–15 cm diameter) calcareous concretions occurred at 0.6–0.9 m in pits dug at the northern end of the Catchment.

The dry bulk density of the upper profile, averaged over five sites, was 1480 kg m⁻³ at 0.1 m, increasing to 1630 kg m⁻³ at 1.0 m.

Vegetation

The vegetation on the Catchment consists of two distinct communities — a mallee community and a heath community as shown in Fig. 1. The mallee community with trees up to 4 m tall covers some 65% of the Catchment area with the dominant species being *Eucalyptus pileata* (Blakely) and *E. eremophila* ([Diels] Maiden). The area of canopy projection by the

mallee canopy was 21% of the ground surface. The heath vegetation is generally 1–2 m tall with the major species being *Melaleuca cuneata* (Turcz.), *M. platycalyx* (Diels.), *M. uncinata* (R.Br.), *Verticordia brownii* (DC.), *V. pressii* (Schau.), *V. roei* (Endl.) and several *Acacia* spp. which could not be identified due to lack of flowering specimens.

Measurement details

Rainfall and runoff

A stainless steel, parabolic, 30°, vee notch, knife edge weir was installed at the Catchment outlet in 1969 to measure runoff. The weir nappe is characteristically aerated and accurate at flow rates above 1×10^{-4} m³ per s. Flow was calculated from stage height in the stilling well using a calibration curve.

Overland flow was also measured from four 4 m² plots under mallee trees and four around heath vegetation during 1981 and 1982. The overland flow plots were 2.24 m diameter, surrounded by a 1.2 mm thick stainless steel ring driven 0.05 m into the soil with 0.25 m above-ground. Overland flow was collected in 62 l drums. Rainfall was measured with a 0.2 mm tipping bucket pluviograph placed near the weir. Additionally, three standard rain gauges were placed on the Catchment boundary and the accumulated rainfall read at least monthly.

Soil moisture

Single neutron moisture meter access tubes were installed to 5 m depth at Sites 1–8 (Fig. 1). The neutron probe was calibrated *in situ* at each site and soil moisture profiles measured monthly from June 1978.

Soil hydraulic conductivity

Saturated hydraulic conductivity was measured at four sites (1, 3, 4, 5; Fig. 1) using 0.3 m diameter infiltration cylinders (Talsma 1969) in nests of four. At Site 3 where the B horizon had a solonchic domed structure the conductivity of the domes was measured using 0.1 m infiltration cylinders.

Soil moisture radials

On 2 July 1980, soil samples were taken from two opposing radials from a single tree of *E. pileata*. Samples were taken at 0.5 m radial intervals, beginning at the base of the tree at depths of 0–0.1 m and 0.1–0.2 m and soil moisture was determined gravimetrically. On 6 August 1980 a further five *E. pileata* trees were similarly sampled.

Dye tracing

To study soil water flow pathways associated with mallee vegetation rhodamine-B (tetra-ethyl rhodamine) dye powder was sprinkled on the soil surface around, and over the crown of, two trees (*E. pileata*) on 1 June 1980.

During October 1980, rhodamine-B dye solution (approx 1 g l^{-1}) was ponded to a depth of 0.1 m in 4 m^2 infiltrometer rings at three sites; two around the bases of mallees (*E. pileata*) and one around a mallee/heath association.

The five sites were excavated to a depth of 1.8 m in April 1981, using a back-hoe.

Rainfall, stemflow and throughfall

Rainfall was measured by four gauges placed in open intergrove areas near the site and the average of the four gauges assumed to represent the rain falling on all trees.

Stemflow was measured on four trees (three *E. pileata* and one *E. eremophila*) from June to October 1981 and from May to November 1982. The four trees had a total of 36 stems and measurements were made by fitting 2 l plastic bottles around each stem about 1 m above ground level. The lower surface of the bottle was sealed around the stem with silicone sealant and at the top a gap of about 10 mm was left around the stem to allow water flow into the bottle while preventing nearly all throughfall entering. The bottles drained into 20 l containers which were read periodically.

Throughfall was measured on three different *E. pileata* and one different *E. eremophila* during May–November 1982 using four 165 mm diameter plastic funnels suspended about 1 m below the canopy of each tree and draining into 2 l bottles.

Height, projected canopy area, individual

stem diameter at 1 m height and the basal area immediately above the lignotuber were measured on each tree.

Results and discussion

The Catchment water balance

Annual rainfall at the Catchment averaged 376 mm for the 11 year period from 1972 to 1982. Rainfall for the May–October period, considered the growing season for agricultural crops in the area, averaged 226 mm or 60% of the annual average. Runoff from the 265 ha Catchment averaged a mere 67 m^3 per year (0.025 mm). There were only two runoff events of any note, the largest being 350 m^3 , (0.13 mm), from 85.8 mm of rain which fell during 27–29 February 1976. The next largest runoff event was 66 m^3 , (0.025 mm) from 32.5 mm rain during 4–5 January 1978. In 3 calendar years (1972, 1975, 1977) no runoff was recorded, and runoff exceeded 0.005 mm in only 11 months of the 11 year period. The seismic survey and borehole drilling showed that there was no groundwater accumulation under the Catchment, thus there was no detectable drainage beyond the root zone.

Over a period of 11 years, it is reasonable to assume no net change in soil moisture storage and, since runoff was negligible, the water balance for the whole Catchment essentially becomes:

Rainfall = interception + evapotranspiration.

Internal redistribution of water

Although runoff from the whole Catchment was negligible, there was considerable redistribution of water within the Catchment. Overland flow as high as 23.4% of rainfall was measured from isolated 4 m^2 plots encompassing heath and mallee within the Catchment (Table 1).

From Table 1 it appears that there was considerably more runoff from around the heath than from around the mallee, with the ratio between the two varying between 2 and 27. The mallee trees redirect up to 30% of the incident rain on the canopy down the stems. Some proportion of this enters the soil at the

TABLE 1. Overland flow from isolated 4 m² plots encompassing heath and mallee vegetation for selected rainfall events during 1981

Date	Rainfall (mm)	Runoff				Heath/mallee ratio
		Mallee mm	Mallee % of rainfall	Heath mm	Heath % of rainfall	
27 May	29.5	2.25	7.6	4.75	16.1	2.1
3 June	30.9	7.70	23.4	*	*	*
2 July	9.8	0.33	3.4	0.63	6.4	1.9
22 July	29.0	2.60	8.9	5.55	19.1	2.1
29 July	8.0	0.03	0.4	0.81	10.1	27.0
19 Aug	34.6	0.70	2.0	4.95	14.3	7.1

*Total not known as collectors were flooded.

base and thus does not contribute to overland flow. Overland flow generated from partial areas within the Catchment infiltrates quickly, and observations of the fate of rhodamine-B dye powder spread on and around the vegetation showed that the maximum distance of overland flow was about 25 m.

It is notable that on none of the dates in Table 1 did runoff from the whole Catchment exceed 0.01 mm. Therefore, overland flow from 4 m² was more than 700 times the runoff from the whole Catchment.

Soil moisture

Soil moisture profiles were measured either monthly or twice monthly so the determination of rates of moisture penetration was limited. A significant feature of all soil moisture profiles was an increase in soil moisture at depth in the hardpan which has a higher bulk density (2000–2100 kg m⁻³) than the mottled clays above (1800–1900 kg m⁻³). The hardpan may impede water movement and would certainly restrict root penetration to zones of weakness. Consecutive monthly soil moisture profiles, as shown in Fig. 2 and 3 under mallee and heath, respectively, show that there was a net increase in soil moisture to at least 5 m under mallee. Under the heath, the profiles are essentially coincident below 3.5 m, indicating either very rapid drainage or no soil moisture recharge. In view of the low hydraulic conductivity of the soil (0.15×10^{-6} m s⁻¹; at saturation 0.48 cm³ cm⁻³), it is highly unlikely that recharge occurred at depth and hence there was very little transfer of water into the 3.5–5.0 m region of the profile. Between 4 June and 1 July 1981 there was 50.5 mm of rainfall and the net

change in soil moisture between 0.5 m and 4.5 m under mallee was 32.8 mm, while under heath it was only 5.4 mm. The rainfall not accounted for in this section of the profile either ran off, evaporated, was stored in the top 0.5 m or drained deeper into the profile. As indicated in Table 1, overland flow would be higher around the heath than around the mallee and may account for 10–16% of the rainfall, which represents 5–8 mm of the total for the period. Evaporation and storage in the upper 0.5 m account for the rest since, as explained above, deep recharge of soil moisture is unlikely. Under the mallee, the soil moisture profiles converge at some depth greater than 5 m and thus there was some storage increase at these depths.

During the summer period from 6 January to 24 February 1982 there was 71.9 mm of rain of which 60.5 mm fell during the 2 days of 21–22 January. Of this input, 58 mm was accounted for as increased soil moisture storage under mallee, whereas less than half the increase (31 mm) was observed under heath for the period (Figs 2, 3).

Both the wetting and drying profiles indicate that roots influence the gain and loss of soil water to considerable depth. It appears that heath roots have their greatest influence in the upper 1 m, with a lesser impact below 3 m. Mallee roots appear to extend their influence the full 5 m, with indications for deeper extraction. During drilling of a bore on the Catchment 150 m south of neutron probe Site 3 in the same mallee community, root material was noted in the drill core at 28 m (M. Martin, Geological Survey of Western Australia, pers. comm. 1984).

At Site 3, the B horizon has a domed

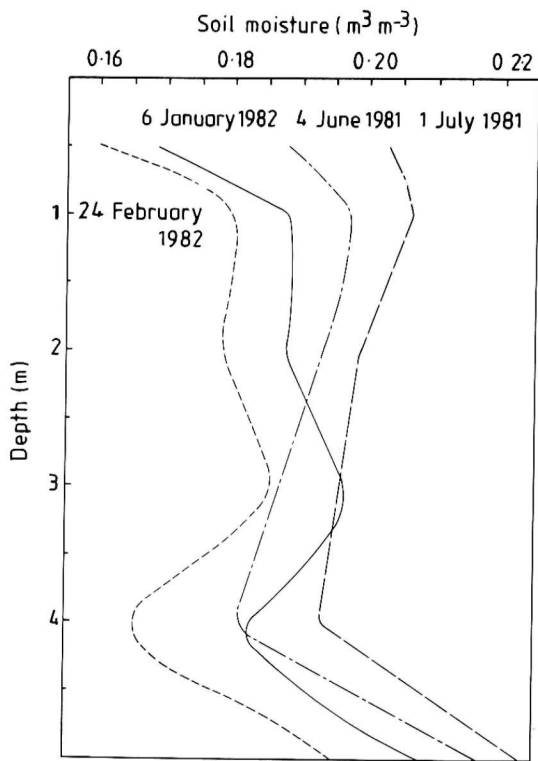


FIG. 2. Soil moisture profiles under mallee. Profiles taken at dates indicated.

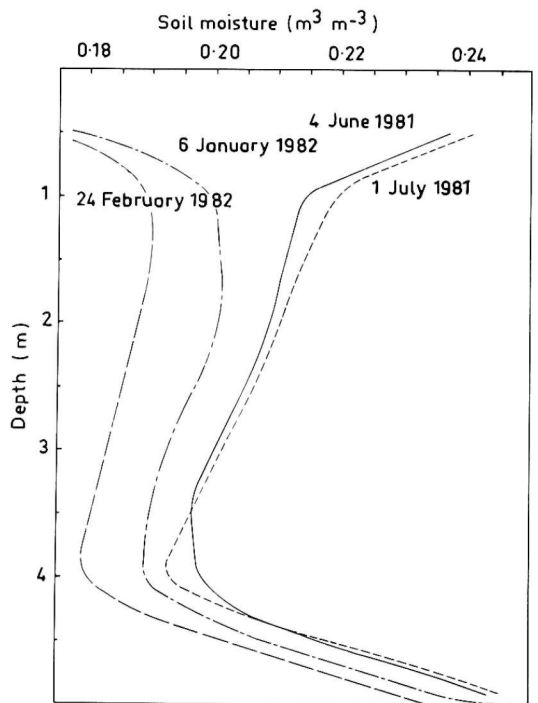


FIG. 3. Soil moisture profiles under heath. Profiles taken at dates indicated.

structure with the clay domes separated by sand filled crevices up to 10 mm wide. Saturated hydraulic conductivity (K_s) was determined on an area containing both domes and crevices and on the domes only. There was no significant difference in the steady state K_s whether the crevices were included or not.

The K_s of 0.15×10^{-6} m per s is not high enough to account for the apparent rate of moisture transmission under the mallee even if flow were assumed to be saturated.

Allison and Hughes (1983) suggested that water flowed down the soil-root interface under

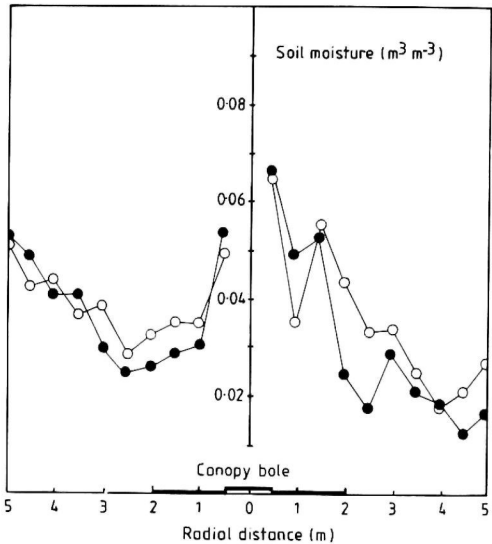


FIG. 4. Soil moisture radials for 0–0.2 m depth from *E. pilcata* tree. Sampled on 2 July 1980.

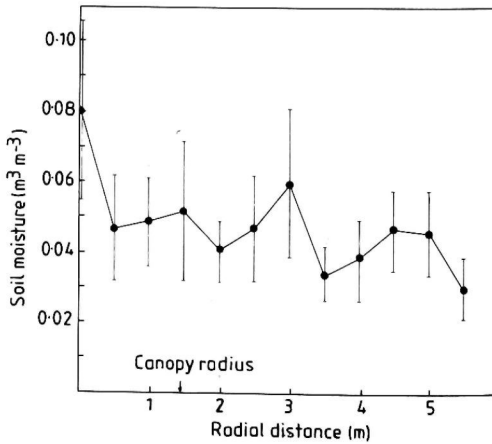


FIG. 5. Mean soil moisture for 0–0.2 m depth sampled on six radii from *E. pilcata* trees on 6 August 1980. Bars represent s.e.m.

mallee vegetation and if this were so it may account for the rapid penetration observed at this site.

Soil moisture radials

Figure 4 shows soil moisture radials for the trees sampled on 2 July 1980 and Fig. 5 depicts the mean soil moisture radials sampled on 6 August 1980. Significantly higher moisture was only detectable immediately at the base of the tree. Once away from the base there was no systematic trend, even at the edge of the canopy where some additional moisture from leaf drip might have been expected.

Dye tracing

The movement of dye applied on 1 June 1980 was visually assessed on 2 July and 6 August 1980. This could bias the results in that low dye concentrations would be masked by the background soil colour and thus not be detected. The dye appeared to be uniformly distributed throughout the A horizon but the only observable penetration into the sandy clay B horizon was associated with pathways around living roots, as shown in Fig. 6. All roots in the B horizon were stained, with dye most obvious within the bark and on the outer phloem surface. Dye was also present in the soil immediately surrounding the root but only moved laterally in areas of relatively high permeability such as in a calcareous module. This is illustrated in Fig. 7 where the dye has stained the nodule moving up to 0.1 m laterally from the root channel. Despite the obvious presence of macrostructural voids, such as interpedal cracks, voided root channels and

channels occupied by dead root material, there was no evidence of dye movement down these pathways. Whether or not these pathways were occluded at the top of the B horizon was not determined. However, since they were not active even under ponded infiltration conditions, occlusion is suggested.

Stemflow, throughfall and interception

Stemflow and throughfall measurements were taken on trees ranging in height from 2.7 m to 4.3 m, with projected canopy areas ranging from 1.8 to 7.8 m² (Table 2). Exposures of the trees varied, with some (e.g. Tree B) being sheltered by surrounding trees, while others (e.g. Trees A and C) were well exposed. Tree dimensions are summarized in Table 2.

Total stemflow for the four trees (A–D) averaged 32% of canopy rainfall over the two seasons of measurement. There was considerable variation between trees and between rainfall events (Table 3). Inter-tree variation could be partly attributed to tree exposure, with mean stemflow least on the small, sheltered Tree B and greatest on the large, well exposed Trees A and C. No comment can be made on the relationship between individual rainfall events and stemflow because of the long intervals over which the stemflow was measured. The intervals usually included several rainfall events of different intensity and duration.

For each measurement period there was a highly significant relationship ($r = 0.90$) between stemflow and stem circumference. An example of the relationship is given in Fig. 8 for

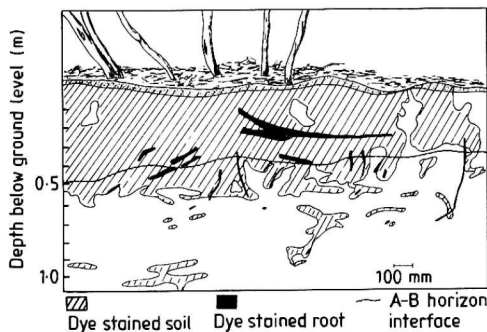


FIG. 6. Rhodamine-B dye in soil under a mallee tree.

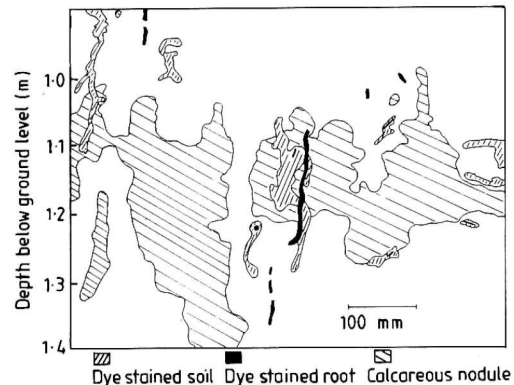


FIG. 7. Detail of distribution of rhodamine-B dye around roots under a mallee tree.

the period 15–16 September 1982. These relationships, one for each period, were used to estimate the stemflow on the individual stems of trees used for measuring throughfall, since the throughfall collectors would have interfered with stemflow measurements. The measured mean throughfall for Trees E–H was 60% of rainfall. Mean stemflow was estimated at 25% and thus, on average, 15% of rainfall was lost through interception storage (Table 4).

Sub-surface water flow

Evidence of the influence of the aerial parts of plants on eventual sub-surface distribution of water has been largely circumstantial. The soil moisture patterns described by Specht (1957) under Dark Island heath after 24 mm of rainfall indicate some influence of the aerial vegetation. Specht highlighted the difference between two vegetative types and suggested that water

TABLE 2. Dimensions of trees used for stemflow (Trees A–D) and throughfall (Trees E–H) measurements

Tree	Species	Height (m)	Projected canopy area (m ²)	Bole area (m ²)	Stems (n)	Total stem circumference (m)	Stem circumference/canopy area (m ⁻¹)
A	<i>E. pileata</i>	4.17	7.8	0.53	14	1.42	0.18
B	<i>E. cremophila</i>	2.85	1.8	0.21	5	0.36	0.20
C	<i>E. pileata</i>	4.30	4.5	0.20	8	0.86	0.19
D	<i>E. pileata</i>	3.55	3.7	0.38	9	0.73	0.20
E	<i>E. pileata</i>	3.57	5.7	0.52		0.77	0.13
F	<i>E. cremophila</i>	2.72	3.4	0.36		0.64	0.19
G	<i>E. pileata</i>	3.93	6.0	0.65		0.78	0.13
H	<i>E. pileata</i>	4.20	2.9	0.25		0.76	0.26

TABLE 3. Total tree stemflow as a percentage of canopy rainfall for Trees A–D

Rainfall period (dates)	Rainfall (mm) for period	Tree A	Percentage stemflow		
			Tree B	Tree C	Tree D
1981					
4–11 June	34.0	39	27	40	30
11–23 June	9.6	39	23	41	33
23 June–2 July	9.9	40	24	42	29
2–22 July	29.0	41	25	49	33
22–29 July	7.6	36	24	35	32
29 July–5 Aug	7.4	29	17	32	26
5–19 Aug	34.6	38	28	41	32
19 Aug–7 Oct	23.6	43	27	42	35
1982					
12–31 May	14.7	22	9	17	16
31 May–16 June	35.5	35	19	33	27
16 June–7 July	12.2	41	19	44	37
7–27 July	19.4	34	19	33	26
27 July–11 Aug	29.1	46	26	47	37
11–26 Aug	35.0	48	24	40	38
26 Aug–15 Sept	15.8	41	24	48	32
15–16 Sept	4.9	35	23	39	29
16–29 Sept	9.8	26	15	26	19
29 Sept–13 Oct	25.8	46	29	48	35
13 Oct–15 Nov	10.2	38	23	39	30
$\bar{x}$		38	22	39	30
s.d.		6	5	8	6

entered the ground near the trunk of *Banksia ornata*, but flowed down between the buried leaf bases of *Xanthorrhoea australis*. Pressland (1976) showed that soil moisture content decreased radially from the base of *Acacia aneura* and his infiltration data show that stemflow could be accommodated by the soil close to the base of the tree.

The individual radial soil moisture data in Fig. 4 and the mean data in Fig. 5 confirm the findings of Pressland (1976), and suggest that stemflow enters the soil at the base of the mallee

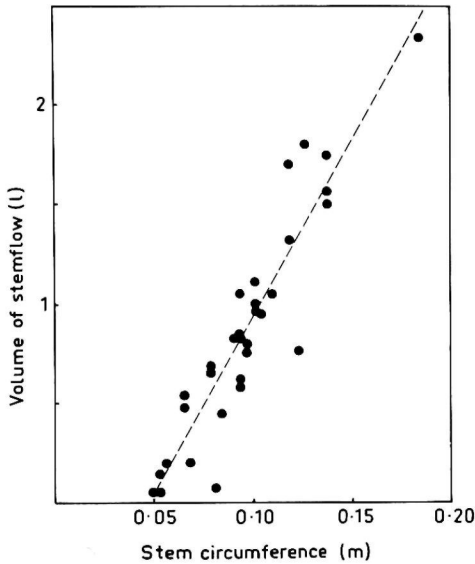


FIG. 8. Relationship between stem circumference and volume of stemflow for the period 15–16 September 1982.

trees. Definitive evidence of the efficacy of the process is the dye tracing (Figs 6,7). In the B horizon, dye was only found in association with living roots. While this is evidence of a direct input of water along the soil-root interface from stemflow, it should not be inferred that water does not penetrate the B horizon via other pathways. Rhodamine-B dye is cationic and thus strongly adsorbed on clay exchange sites. It may be that dye moving in other pathways, associated with small quantities of water, was adsorbed near the top of the B horizon.

Not all of the stemflow enters the ground near the bole. Overland flow from the 4 m² around the bole of a mallee tree was as high as 23% of area rainfall. This extreme value was measured in only one event — all other measurements were less than 10% of rainfall. So the rate of arrival of water at the bole can exceed the total infiltration capacity of the soil and the soil-root interface which contrasts with Pressland's (1976) finding and is probably indicative of different soil types.

Rainfall partitioning above-ground

The spatial arrangement of stemflow collectors and throughfall collectors precluded measurement of both on the same tree. It was necessary to estimate stemflow from the stemflow-stem circumference relationship for single stems and this introduced some possible errors. The stemflow-stem circumference relationship substitution was found to be valid for trees with a similar total stem circumference to canopy

TABLE 4. Throughfall, stemflow and interception by Trees E–H for period 12 May to 15 November 1982

Measurement period (dates)	Rainfall (mm)	Means of Trees E–H		
		Measured throughfall (%)	Estimated stemflow (%)	Estimated interception (%)
12–31 May	14.7	63	13	24
31 May–16 June	35.5	61	23	16
16 June–7 July	12.2	51	31	18
7–27 July	19.4	69	22	9
27 July–10 Aug	29.1	62	30	8
10–25 Aug	35.0	55	29	16
25 Aug–15 Sept	15.8	57	28	15
15–16 Sept	4.9	60	25	15
16–29 Sept	9.8	56	17	27
29 Sept–13 Oct	25.8	73	31	(–4)
13 Oct–15 Nov	10.2	54	26	20
$\bar{x}$		60	25	15
s.d.		6	6	8

area ratio (SC/CA). If SC/CA of the object tree is lower than that of the calibration tree then stemflow will be underestimated. Only Tree F had an SC/CA within the range of the trees used for stemflow measurement (Table 2). The mean SC/CA for the object trees (E–H) was 7% lower than for the measured trees (A–D). Thus, the stemflows given in Table 4 may well be underestimates. This would result in an overestimate of interception.

Considering the above possible errors, the mean interception loss of 15% compares favourably with the 13% found by Pressland (1973) on *Acacia aneura*. Both Slatyer (1965) and Pressland (1973) found that interception loss was limited to 1.5–2.5 mm per storm event. In this study it was not possible to determine the individual storm limit of interception loss.

The anomalous negative interception value for the period 29 September to 12 October 1982 (Table 4) may be due, in part, to the ‘parachute’ shape of the trees which, during rain events accompanied by high wind velocities, exposes an area to incident rainfall greater than the projected canopy area. Thus, there would be an underestimate of the rain actually falling on the canopy. This discrepancy would increase as the incident angle of rainfall decreased with increasing wind velocity. The period of the anomalous measurement was a period of relatively high winds with the mean velocity for the 13 days being 2.4 m s^{-1} .

The effect of stemflow is to increase the potential for recharge of soil moisture near the base of the tree. If it is assumed that the stemflow infiltrates around the bole in an

annulus equivalent in area to the bole, then an effective basal precipitation can be calculated using the canopy area/bole area ratio.

The ratio between effective basal ‘precipitation’ and actual precipitation can be greater than 10. Table 5 presents the calculated effective basal precipitation for the four trees on which stemflow was measured during 1981. The variation in mean percentage effective basal precipitation essentially reflects the difference in the canopy area/basal area ratio of the individual trees.

Impact of rainfall redistribution by vegetation on Catchment hydrology

While the stemflow and interception constitute some 40% of the rain falling on the canopy, the impact of this appears small when the projected ground cover of 21% is accounted for. The net effect is that the tree canopy and stems only redistribute about 8% of the rain falling on the Catchment. However, small changes in gross hydrology can have very significant consequences. For example, in southern Western Australia, clearing land and developing for agriculture results in an additional 4–10% of mean annual rainfall percolating beyond the root zone to recharge groundwaters (George 1978; Peck & Hurlle 1973) which, after 15–20 years, causes secondary salinity in lower landscape areas.

Mallee vegetation appears to be adapted to this semi-arid environment. The canopy collects water and channels it down the stem. The water then flows down the soil-root interface

TABLE 5. Effective basal precipitation due to stemflow*

Canopy rainfall (mm)	Tree A		Tree B		Tree C		Tree D	
	mm	%	mm	%	mm	%	mm	%
34.0	194	571	77	227	306	901	99	291
9.6	55	571	19	194	89	927	31	321
9.9	59	592	19	194	94	954	32	321
29.0	177	612	61	211	322	1113	93	321
7.6	40	530	16	211	40	530	28	367
7.4	32	428	11	146	53	715	18	245
34.6	198	571	84	243	321	928	106	306
23.6	149	632	54	227	219	928	79	337
Mean %		563		207		874		314

*Percentages are the increase of effective basal precipitation over the canopy rainfall during rainfall periods in 1981.

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where it is stored at depths as great as 28 m for use during the dry summer.

Removal of the mallee vegetation would obviously affect the redistribution of rainfall and, although the components that are affected are a small proportion of the rainfall, the time-integrated effect can be very significant. Under mallee, water penetration deep into the soil profile is via preferred pathways. Whether or not these pathways operate after removal of the mallee is unknown. However, the major component enabling these pathways to operate is the stemflow, which maintains saturated conditions where the roots enter the dense B horizon. With the removal of the trees resulting in a reduction in stemflow, it is unlikely that the pathways will operate as effectively as when they were fed stemflow water. Hence, clearing may change the mode of entry of water into the deep profile from a largely preferred pathway system to a more uniform flow system. This requires further investigation.

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